

SiteSpeak 2.0

Autonomous Industrial Safety Voice Agent & Digital Twin



Hands-Free Voice Agent

Powered by AssemblyAI Voice Agent API (Universal-3 Pro STT, ultra-fast turn-taking, and managed LLM).



Interactive Digital Twin

Real-time SVG schematic dynamically pinpoints hazards, spill locations, and emergency equipment.



Certified OSHA Form 301

Instant transformation of natural field conversations into legally certified compliance reports.

Paperwork Takes 20 Minutes. Toxic Spills Don't Wait.

\$170 Billion+

ANNUAL US WORKPLACE INJURY COSTS (OSHA / NSC)

Severe delays in reporting toxic chemical leaks, fires, and equipment failures multiply damage exponentially.

🧤 The "Dirty Gloves" Paradox

Workers in chemical plants, construction zones, and logistics facilities wear heavy PPE gloves. They cannot type on smartphones, touchscreens fail in grease and rain, and paper logs get filed hours late.

✗ Legacy Failure Mode

- 20+ minute average delay to notify plant safety managers.
- SDS chemical sheets buried in three-ring binders 300 yards away.
- Vague radio calls: "Hey, there's a leak over by building B!" with no exact column or containment details.

✅ The SiteSpeak Solution

- **Hands-Free:** Worker presses one big button or speaks naturally.
- **Sub-Second Action:** Instantly fetches HAZMAT PPE requirements, alerts evacuation zones, and sounds emergency alarms.

Speak Naturally. The Agent Takes Command.

A natural 90-second conversation captures every fact, updates databases, and sounds alarms in real-time.

1

Spoken Description

"Chemical spill near Bay 3! One person slipped, and there's a strong ammonia vapor smell!"

Universal-3 Pro transcribes streaming audio at 24 kHz PCM16 with full VAD barge-in.

2

10 Real-Time JSON Tools

Agent fires JSON-schema tools sub-second:

- `lookup_site("bay 3")`
- `assess_hazmat_protocol("ammonia")`
- `create_incident_draft`

3

Evacuation & Compliance

The agent speaks aloud:

"Advising immediate 100-meter perimeter and Level B vapor suits. Nearest eyewash is at Column B2. Is anyone injured?"

AssemblyAI Voice Agent API Architecture

[ Field Worker Mic] | (24 kHz PCM16 WebSocket) ▼
[ AssemblyAI Voice Agent API] |— Universal-3
Pro STT |— Real-Time VAD & Barge-In |— Managed
LLM with 10 JSON Tools | | tool.call / tool.result
▼ [ SiteSpeak 2.0 Client (React 18 + Vite)] |—
Field Console & Waveform Visualizer |— 2D/3D
Digital Twin Safety Schematic |— Supervisor
Operations Center | ▼ (REST API) [ SiteSpeak
Backend (Fastify + SQLite WAL)] |— Single-Use
Token Minter |— 10 Tool Handlers & Audit Trail |—
OSHA Standard 1904 Form 301 Engine

Enterprise Token Security

API keys never touch the client. Backend mints single-use, 300-second session tokens via `agents.assemblyai.com/v1/token`.

Bidirectional Tool Execution

Client executes tools against the Fastify backend, persisting every query and state transition in SQLite with sub-millisecond audit metrics.

Seamless Barge-In Turn Taking

Workers can interrupt mid-sentence during emergency escalations without losing previously captured report fields.

Beyond Chatbots: Real Operational Execution

Life-Safety & HAZMAT Tools

`assess_hazmat_protocol`

→ SDS UN numbers, Level A/B/C PPE, evacuation radii.

`find_emergency_equipment`

→ Column coordinates for spill kits, eyewashes, AEDs.

`trigger_emergency_broadcast`

→ Sounds site siren and triggers evacuation banner.

`query_site_status`

→ Bidirectional spoken safety briefings.

Compliance & Incident Lifecycle

`lookup_site`

→ Resolves "North Yard Bay 3" to canonical zone.

`create_incident_draft`

→ Initializes draft with unique incident ID.

`update_incident`

→ Patches injuries, severity, and zone dynamically.

`add_followup`

→ Assigns EMT, HAZMAT, or Maintenance teams.

`file_incident & cancel_incident`

→ Validates required fields and seals the report.

Visual Grounding + OSHA 301 Compliance



Interactive SVG Digital Twin

- Live multi-zone floorplans for North Yard and Warehouse B.
- Glowing, pulsing hazard markers dynamically appear at the spoken coordinates.
- Safety equipment overlays (Eyewash, Spill Kits, AEDs, Fire Extinguishers) with exact distance indicators.
- Dynamic evacuation perimeter radius rendering.



Certified OSHA Form 301

- Formatted to official ****OSHA Standard 1904**** requirements.
- Maps voice transcript facts directly into regulatory fields: injury mechanics, object/substance involved, treatment provided.
- 1-click printable modal and JSON regulatory export for compliance audits.

Quantifiable Enterprise Value

92%

REPORTING TIME REDUCTION

From 20-minute manual paperwork down to a 90-second natural voice conversation.

< 1 sec

HAZMAT RESPONSE LATENCY

Immediate SDS chemical protocols spoken directly to first responders before entering danger zones.

100%

OSHA AUDIT READINESS

Zero missing fields, automated timestamps, complete millisecond tool audit trail.

SiteSpeak 2.0 — Safe Sites Speak Clearly

What Makes SiteSpeak Unique

- **Deep Tech Integration:** 10 dynamic tools on AssemblyAI Voice Agent API.
- **Simulation Studio:** 6 automated multi-turn industrial scenarios for zero-mic testing.
- **Complete Solution:** Field Voice Console, Digital Twin, HAZMAT Compass, and OSHA 301 Engine.

Future Roadmap

- Wearable BLE Bone-Conduction Earbuds for noisy plants.
- Computer Vision Camera integration for automated hazard verification.
- Multilingual turn-taking for international construction crews.

Thank you, AssemblyAI & Lablab.ai Judges!